Monte Carlo Coulomb Collisions for Guiding-Centre Particles in Boozer Coordinates: Implementation Notes

firm3d — `ep_thermal_collisions`

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Contents

1 Overview

These notes describe the algorithm used in `trace_particles_boozer_with_collisions` to include stochastic Coulomb collisions for an energetic particle (EP) tracing through one or more Maxwellian background species.

The approach couples the deterministic guiding-centre (GC) orbit equations in Boozer coordinates with a Milstein stochastic term for the velocity-space diffusion. The full state vector is $(s, \theta, \zeta, v, \xi)$, where $v = |\boldsymbol{v}|$ is the total speed and $\xi = v_\parallel / v$ is the pitch.

Primary references:

- Hirvijoki et al. [1] — Monte Carlo implementation of the guiding-centre Fokker–Planck equation in (v, ξ) coordinates; Eqs. (32)–(33) of that paper define the SDE system used here.
- Hirvijoki et al. [2] — ASCOT4/5 code description; source file `mccc_coefs.h` defines the Coulomb logarithm convention adopted here.
- Boozer & Kuo-Petravic [4] — pitch-angle representation in Boozer coordinates.

2 Guiding-Centre Equations in Boozer Coordinates

In Boozer coordinates (s, θ, ζ), where s is the normalised toroidal flux, θ the poloidal angle, and ζ the toroidal angle, the vacuum GC equations are

$$\dot{s} = -\frac{m\,v^2(1+\xi^2)}{2\,q\,\psi_0}\,\frac{\partial B}{\partial \theta}\,\frac{1}{B}, \tag{1}$$

$$\dot{\theta} = \frac{m\,v^2(1+\xi^2)}{2\,q\,\psi_0}\,\frac{\partial B}{\partial s}\,\frac{1}{B} + \frac{\iota\,\xi\,v\,B}{G}, \tag{2}$$

$$\dot{\zeta} = \frac{\xi\,v\,B}{G}, \tag{3}$$

$$\dot{\xi} = -\frac{(\iota\,\partial_\theta B + \partial_\zeta B)\,v\,(1-\xi^2)}{2G} \quad \text{[mirror force]}, \tag{4}$$

where ψ_0 is the reference poloidal flux, $G(s)$ is the covariant toroidal field component, and $\iota(s)$ is the rotational transform. The magnetic moment $\mu = v^2(1-\xi^2)/(2B)$ is reconstructed locally at each evaluation.

With collisions the equations for v and ξ acquire deterministic drift terms (Section 4) and stochastic noise terms (Section 5.2).

3 Background Species and Profiles

Each background species b is described by a `ThermalBackground` object specifying:

- $n_b(s)$ — number density [m^{-3}], callable of the flux label $s \in [0, 1]$;
- $T_b(s)$ — temperature [J], callable (use `T_keV` $\times$ `1e3` $\times e$);
- m_b — species mass [kg];
- q_b — species charge [C] (signed).

Internally the profiles are pre-evaluated on a uniform s-grid of 512 points and linearly interpolated in C++.

4 Collision Coefficients

Notation

The symbols used here follow the ASCOT5 source code (`mccc_coefs.h`, [3]); Table 1 shows the correspondence with Hirvijoki et al. [1], whose Eqs. (32)–(33) define the SDE structure.

This paper	ASCOT5 `mccc_coefs.h`	Hirvijoki et al. [1]
$D_\parallel$	`Dpara`	$D_k/2$
ν_D	`nu` $= 2D_\perp/v^2$	$2D_\perp/v^2$
Q	`Q`	(Einstein-relation drag in K)
K	`K` $= Q + D'_\parallel + 2D_\parallel/v$	K (drift in Eq. 32)

Table 1: Notation correspondence. The noise amplitude for v is $g_v = \sqrt{2D_\parallel}$ (this paper) $= \sqrt{D_k}$ (Hirvijoki 2013).

4.1 Debye Length

The total Debye screening length from all background species is [5, 6]

$$\frac{1}{\lambda_D^2} = \sum_b \frac{n_b\, q_b^2}{\varepsilon_0\, T_b}. \tag{5}$$

4.2 Coulomb Logarithm

The Coulomb logarithm is defined as the ratio of maximum to minimum impact parameters [5, 6, 7]:

$$\ln\Lambda_{ab} = \ln\left(\frac{\lambda_D}{b_{\min}}\right), \qquad b_{\min} = \max(b_{\mathrm{cl}},\, b_{\mathrm{qm}}). \tag{6}$$

The maximum impact parameter is the Debye length (Eq. (5)). The minimum impact parameter is the larger of the classical 90°-deflection distance and the quantum-mechanical de Broglie wavelength of the reduced mass:

$$b_{\rm cl} = \frac{|q_a\, q_b|}{4\pi\varepsilon_0\, m_r\, \bar{v}^2}, \tag{7}$$

$$b_{\rm qm} = \frac{\hbar}{2\, m_r\, \sqrt{\bar{v}^2}}, \tag{8}$$

where $m_r = m_a m_b/(m_a + m_b)$ is the reduced mass.

The effective velocity scale $\bar{v}^2$ is taken as

$$\bar{v}^2 = v^2 + v_{{\rm th},b}^2, \qquad v_{{\rm th},b} = \sqrt{2T_b/m_b}, \tag{9}$$

following the ASCOT5 source convention [3]. This form is a numerical interpolation device, not derived from first principles: Ichimaru [7] derives $b_{\rm cl}$ and $b_{\rm qm}$ only in the fast-particle limit $v \gg v_{{\rm th},b}$ (Section 4.3B of that reference), where $\bar{v} \approx v$. The $\bar{v}^2$ form extends the formula to the slow-particle regime without a discontinuous switch: fast EP against ions ($v \gg v_{{\rm th},b}$, $\bar{v} \approx v$) and slow EP against electrons ($v \ll v_{{\rm th},e}$, $\bar{v} \approx v_{{\rm th},e}$).

The quantum branch $b_{\rm qm} > b_{\rm cl}$ applies when

$$\frac{2\pi\varepsilon_0\, \hbar\, \sqrt{\bar{v}^2}}{|q_a\, q_b|} > 1, \tag{10}$$

i.e. when the de Broglie wavelength of the reduced mass particle exceeds the classical distance of closest approach. For 3.52 MeV α-particles scattering off 10 keV electrons one finds $b_{\rm qm} \approx 22\, b_{\rm cl}$, so the quantum branch is active throughout alpha slowing-down against electrons. If $\ln\Lambda \le 0$ (Debye length below the minimum impact parameter, e.g. $T_b \to 0$ at finite density) the binary-collision model is undefined and an exception is raised; ASCOT5 reaches the equivalent outcome by aborting markers through its input-evaluation errors, and its input profiles keep the temperature finite wherever the density is nonzero. A warning is issued for $0 < \ln\Lambda < 2$, where the perturbative expansion underlying the Fokker–Planck operator begins to lose validity.

Both checks are applied to the profiles once, before tracing starts, rather than from inside the right-hand side: the latter is evaluated roughly seven times per accepted step per particle, so a per-evaluation warning would emit millions of duplicate lines, and a per-evaluation exception would fire on whichever MPI rank owns the offending particle, hanging the remaining ranks in the collective that gathers results. Since $\ln\Lambda$ increases monotonically with v through $\bar{v}^2$, evaluating at $v \to 0$ bounds it from below over all particle speeds, making the up-front check conservative but never optimistic.

4.3 Chandrasekhar G Function

The Chandrasekhar function [8] is

$$G(x) = \frac{\mathrm{erf}(x) - \dfrac{2x}{\sqrt{\pi}}\, e^{-x^2}}{2x^2}, \qquad x = v/v_{{\rm th},b}. \tag{11}$$

Its derivative satisfies the recurrence

$$G'(x) = \frac{2}{\sqrt{\pi}}\, e^{-x^2} - \frac{2\, G(x)}{x}. \tag{12}$$

Key limiting behaviours:

$$x \to 0: \quad G \to 0, \quad G' \to 2/(3\sqrt{\pi}) \approx 0.376;$$
$$x \to \infty: \quad G \to 1/(2x^2), \quad G' \to 0.$$

4.4 Collision Frequency Prefactor

Define the Rosenbluth prefactor for species b [9, 10]:

$$\Gamma_b = \frac{n_b\, q_a^2\, q_b^2\, \ln \Lambda_{ab}}{4\pi\varepsilon_0^2\, m_a^2}. \tag{13}$$

4.5 Pitch-Angle Scattering Frequency

$$\nu_D^{(b)} = \frac{\Gamma_b\, [\mathrm{erf}(x) - G(x)]}{v^3} \;\equiv\; \frac{2D_\perp^{(b)}}{v^2}. \tag{14}$$

The second form uses the perpendicular diffusion coefficient $D_\perp^{(b)} = \Gamma_b[\mathrm{erf}(x) - G(x)]/(2v)$ and matches ASCOT5's `mccc_coefs_nu` and the $2D_\perp/v^2$ notation of Hirvijoki et al. [1, 2].

4.6 Parallel Diffusion Coefficient

$$D_\parallel^{(b)} = \frac{\Gamma_b\, G(x)}{v}, \qquad \frac{\partial D_\parallel^{(b)}}{\partial v} = \frac{\Gamma_b}{v}\left(\frac{G'(x)}{v_{\mathrm{th},b}} - \frac{G(x)}{v}\right). \tag{15}$$

In Hirvijoki et al. [1] the parallel diffusion coefficient is denoted D_k, with $D_\parallel = D_k/2$; the noise amplitude for the speed SDE is $\sqrt{D_k} = \sqrt{2D_\parallel}$ in both conventions. ASCOT5 names this coefficient `Dpara`. The v-derivative is required by the Milstein correction term (Section 5.2). Note that it differentiates Γ_b as though $\ln \Lambda_{ab}$ were constant, even though the $\ln \Lambda_{ab}$ actually evaluated depends on v through $\bar{v}^2$; this follows ASCOT5 and has a small but systematic consequence for the equilibrium, quantified in Section 10.

4.7 Deterministic Drag in v

The drag coefficient (ASCOT5: `mccc_coefs_Q` [3]) is fixed by the Einstein relation:

$$Q^{(b)} = -\frac{m_a\, v}{T_b}\, D_\parallel^{(b)} = -\Gamma_b\, \frac{m_a\, G(x)}{T_b}. \tag{16}$$

The total deterministic drift in speed is

$$K^{(b)} = Q^{(b)} + \frac{\partial D_\parallel^{(b)}}{\partial v} + \frac{2\, D_\parallel^{(b)}}{v} \tag{17}$$

(ASCOT5's `mccc_coefs_K` [3]; the drift coefficient K of Eq. (32) in Hirvijoki et al. [1]).

Itô transformation leading to Eq. (17)

Equation (17) can be obtained two independent ways, which cross-check each other.

(i) Itô's formula on $v = |\boldsymbol{v}|$. In three-dimensional velocity space the collisional motion of the test particle is

$$\mathrm{d}\boldsymbol{v} = F_{\mathrm{dyn}}\,\hat{\boldsymbol{v}}\,\mathrm{d}t + \boldsymbol{\sigma}\cdot\mathrm{d}\boldsymbol{W}, \qquad \boldsymbol{\sigma}\boldsymbol{\sigma}^{\top} = 2D_{\parallel}\,\hat{\boldsymbol{v}}\hat{\boldsymbol{v}} + 2D_{\perp}\,(\mathsf{I} - \hat{\boldsymbol{v}}\hat{\boldsymbol{v}}), \tag{18}$$

where F_{dyn} is the dynamical friction [8, 10, 11]. The speed $v = |\boldsymbol{v}|$ is a nonlinear function of $\boldsymbol{v}$, so Itô's formula applies with $\nabla_{\boldsymbol{v}}\, v = \hat{\boldsymbol{v}}$ and Hessian $\nabla_{\boldsymbol{v}}\nabla_{\boldsymbol{v}}\, v = (\mathsf{I} - \hat{\boldsymbol{v}}\hat{\boldsymbol{v}})/v$:

$$\mathrm{d}v = \Big(F_{\mathrm{dyn}} + \frac{2D_{\perp}}{v}\Big)\mathrm{d}t + \sqrt{2D_{\parallel}}\,\mathrm{d}W_v. \tag{19}$$

The $2D_{\perp}/v$ drift is the Bessel-process effect: kicks transverse to $\boldsymbol{v}$ do not change v at first order, but they lengthen the vector at second order, $\Delta v = |\boldsymbol{v} + \Delta\boldsymbol{v}_{\perp}| - v \approx |\Delta\boldsymbol{v}_{\perp}|^2/2v > 0$.

(ii) Fokker–Planck matching (detailed balance). The Itô SDE $\mathrm{d}v = K\,\mathrm{d}t + \sqrt{2D_{\parallel}}\,\mathrm{d}W_v$ evolves the speed density $P(v) \propto v^2 f(v)$ according to $\partial_t P = -\partial_v(KP) + \partial_v^2(D_{\parallel}P)$. Zero flux on the equilibrium $P_{\mathrm{eq}} \propto v^2\, e^{-m_a v^2/2T_b}$ requires $KP_{\mathrm{eq}} = \partial_v(D_{\parallel}P_{\mathrm{eq}})$, i.e.

$$K = -\frac{m_a v}{T_b}\,D_{\parallel} + \frac{\partial D_{\parallel}}{\partial v} + \frac{2\,D_{\parallel}}{v}, \tag{20}$$

which is Eq. (17) with Q as in Eq. (16). Here $\partial_v D_{\parallel}$ is the multiplicative-noise (Itô) drift that appears when the diffusion term is commuted into advective form, and $2D_{\parallel}/v$ comes from the Jacobian v^2 of spherical velocity-space coordinates.

Consistency of (i) and (ii). For Maxwellian field particles the dynamical friction obeys the identity

$$F_{\mathrm{dyn}}^{(b)} = -\Big(1 + \frac{m_a}{m_b}\Big)\frac{2x^2\,G(x)\,\Gamma_b}{v^2} = Q^{(b)} + \frac{\partial D_{\parallel}^{(b)}}{\partial v} + \frac{2\,\big(D_{\parallel}^{(b)} - D_{\perp}^{(b)}\big)}{v}, \tag{21}$$

so that $F_{\mathrm{dyn}} + 2D_{\perp}/v = Q + \partial_v D_{\parallel} + 2D_{\parallel}/v = K$ exactly. (F_{dyn} is ASCOT5's `mccc_coefs_F`, which is not used in the guiding-centre scheme.) In the fast-particle limit $x \gg 1$ this identity encodes the classic cancellation

$$F_{\mathrm{dyn}} + \frac{2D_{\perp}}{v} \;\longrightarrow\; -\Big(1 + \frac{m_a}{m_b}\Big)\frac{\Gamma_b}{v^2} + \frac{\Gamma_b}{v^2} = -\frac{m_a}{m_b}\,\frac{\Gamma_b}{v^2}\;:$$

most of the dynamical friction is compensated by transverse diffusion, and the net slowing-down rate is set by the mass ratio. Note in particular that Q is *not* the dynamical friction — it is the Einstein-relation drag $-(m_a v/T_b)D_{\parallel}$.

4.8 Summation over Species

All coefficients are additive over background species:

$$\nu_D = \sum_b \nu_D^{(b)}, \quad D_{\parallel} = \sum_b D_{\parallel}^{(b)}, \quad K = \sum_b K^{(b)}. \tag{22}$$

The Debye length used in Eq. (6) is computed from all species simultaneously (Eq. (5)), prior to the per-species loop.

5 Stochastic Differential Equations

5.1 Full SDE System

The complete SDE for the five-dimensional state $(s, \theta, \zeta, v, \xi)$ is [1]

$$\mathrm{d}s = \dot{s}_{\mathrm{orb}}\,\mathrm{d}t, \tag{23}$$

$$\mathrm{d}\theta = \dot{\theta}_{\mathrm{orb}}\,\mathrm{d}t, \tag{24}$$

$$\mathrm{d}\zeta = \dot{\zeta}_{\mathrm{orb}}\,\mathrm{d}t, \tag{25}$$

$$\mathrm{d}v = K(v,s)\,\mathrm{d}t + \sqrt{2D_\parallel}\,\mathrm{d}W_v, \tag{26}$$

$$\mathrm{d}\xi = \dot{\xi}_{\mathrm{mirror}}\,\mathrm{d}t - \xi\,\nu_D\,\mathrm{d}t + \sqrt{\nu_D(1-\xi^2)}\,\mathrm{d}W_\xi, \tag{27}$$

where $\mathrm{d}W_v$ and $\mathrm{d}W_\xi$ are independent Wiener increments with $\mathbb{E}[\mathrm{d}W^2] = \mathrm{d}t$. Equations (26)–(27) correspond to Eqs. (32)–(33) of Hirvijoki et al. [1], with the identification $D_\parallel \leftrightarrow D_k/2$ and $\nu_D \leftrightarrow 2D_\perp/v^2$ in their notation. The noise amplitude for ξ ensures that the marginal distribution of ξ converges to a uniform distribution on $[-1,1]$ in the isotropisation limit $\nu_D t \gg 1$.

5.2 Milstein Scheme

After each accepted Dormand–Prince (DP) step of physical size h, the stochastic increments $\Delta W_v \sim \mathcal{N}(0,h)$ and $\Delta W_\xi \sim \mathcal{N}(0,h)$ are generated and applied via the Milstein method [12]:

$$v \leftarrow v + g_v\,\Delta W_v + \tfrac{1}{2}\frac{\partial g_v}{\partial v}\big(\Delta W_v^2 - h\big), \tag{28}$$

$$\xi \leftarrow \xi + g_\xi\,\Delta W_\xi + \tfrac{1}{2}\frac{\partial g_\xi}{\partial \xi}\big(\Delta W_\xi^2 - h\big), \tag{29}$$

where the noise amplitudes and their derivatives are

$$g_v = \sqrt{2D_\parallel}, \qquad \frac{\partial g_v}{\partial v} = \frac{\partial D_\parallel/\partial v}{\sqrt{2D_\parallel}}, \tag{30}$$

$$g_\xi = \sqrt{\nu_D(1-\xi^2)}, \qquad \frac{\partial g_\xi}{\partial \xi} = \frac{-\xi\,\nu_D}{\sqrt{\nu_D(1-\xi^2)}}. \tag{31}$$

In terms of the coefficient structs, the Milstein correction for ξ simplifies to

$$\delta\xi_{\mathrm{Milstein}} = -\tfrac{1}{2}\,\xi\,\nu_D\,\big(\Delta W_\xi^2 - h\big). \tag{32}$$

Boundary conditions follow ASCOT5 (`mccc_gc_milstein.c`, [3]). The speed reflects off a thermal cutoff,

$$v < v_{\mathrm{cut}} \;\Rightarrow\; v \leftarrow 2\,v_{\mathrm{cut}} - v, \qquad v_{\mathrm{cut}} = 0.1\,\sqrt{T_{\min}/m_a}, \tag{33}$$

where $T_{\min}$ is the coldest active background species (ASCOT5's `MCCC_CUTOFF` uses its first background species), so that the collision coefficients, which diverge as $v \to 0$, are never evaluated deep below the thermal speed. The pitch mirrors at the boundary: $|\xi| > 1 \Rightarrow \xi \leftarrow \mathrm{sign}(\xi)\,(2-|\xi|)$; a hard clamp would pile up probability at exactly $\xi = \pm 1$.

Order of convergence

Only the diagonal Milstein corrections are retained above. In more than one dimension these alone give order-1 strong convergence just when the diffusion matrix is commutative, i.e. $L^{j_1} g^{k,j_2} = L^{j_2} g^{k,j_1}$ with $L^j = \sum_k g^{k,j}\, \partial_k$ [12]. The diffusion here is diagonal — $\mathrm{d}W_v$ drives v and $\mathrm{d}W_\xi$ drives ξ — so with $k = \xi$,

$$L^v g_\xi = g_v \frac{\partial g_\xi}{\partial v} \neq 0, \qquad L^\xi g_v = g_\xi \frac{\partial g_v}{\partial \xi} = 0, \tag{34}$$

the first being nonzero because ν_D depends on v, and the second vanishing because $D_\parallel$ does not depend on ξ. The condition fails, so the double stochastic integrals (Lévy areas) $I_{(v,\xi)}$ and $I_{(\xi,v)}$ do not cancel and cannot be recovered from $\Delta W_v, \Delta W_\xi$ alone. Dropping them — as ASCOT5 also does — reduces *strong* convergence to order 1/2 in that coupling. *Weak* convergence, which governs the moments and distribution functions a Monte Carlo transport calculation actually reports, remains order 1 and is unaffected.

6 Adaptive Integration Strategy

6.1 Dormand–Prince Solver

The deterministic ODE part (all five components of the drift) is integrated with the classic 4(5)-order embedded Dormand–Prince (FSAL) method [13]. Step-size control uses mixed absolute/relative tolerances:

$$\mathrm{err}_i = \left| y_i^{(5)} - y_i^{(4)} \right| \leq \delta_{\mathrm{abs}} + \delta_{\mathrm{rel}}\, |y_i|. \tag{35}$$

Default tolerances: $\delta_{\mathrm{abs}} = \delta_{\mathrm{rel}} = 10^{-9}$.

6.2 Minimum Step Size

The parameter `DP_hmin` (default `0.0`) sets a lower bound on the physical step size h in seconds. A value of 10^{-10}–10^{-9} s prevents the solver from stalling at bounce points while keeping steps small enough that the orbital dynamics remain accurate.

6.3 Operator Splitting

The SDE is solved via Lie–Trotter operator splitting:

1. **Deterministic half:** advance $(s, \theta, \zeta, v, \xi)$ by one DP step of physical time h using the full drift $(\dot{s}, \dot{\theta}, \dot{\zeta}, K, -\xi\nu_D + \dot{\xi}_{\mathrm{mirror}})$.

2. **Stochastic half:** apply Milstein noise to (v, ξ) at the accepted state, using h as the diffusion window.

The trajectory is saved *before* the Milstein step at each t_{save} checkpoint, so saved states correspond to the deterministic prediction (consistent with the standard output convention).

7 Output Format

Each particle trace is a NumPy array of shape (ntimesteps, 6) with columns

$$[t,\ s,\ \theta,\ \zeta,\ v_\parallel,\ v].$$

From these the kinetic energy $E = \frac{1}{2}mv^2$ and the magnetic moment $\mu = (v^2 - v_\parallel^2)/(2B)$ can be reconstructed at each saved point. When `forget_exact_path=True` only the first and last rows are retained.

Each particle uses RNG seed `rng_seed`$+i$, making multi-particle MPI runs fully reproducible.

8 Python Interface

```
from firm3d.field.collisions import ThermalBackground, \
    trace_particles_boozer_with_collisions
from firm3d.util.constants import (
    PROTON_MASS, ELECTRON_MASS, ELEMENTARY_CHARGE, ONE_EV,
    ALPHA_PARTICLE_MASS, ALPHA_PARTICLE_CHARGE,
    FUSION_ALPHA_PARTICLE_ENERGY,
)

# Typical DT-fusion plasma backgrounds
n0 = 1e20          # m^-3
deuterium = ThermalBackground(
    n_profile=lambda s: n0 * (1 - s),
    T_profile=lambda s: 10e3 * ONE_EV * (1 - s),
    mass=2 * PROTON_MASS,
    charge=ELEMENTARY_CHARGE,
)
tritium = ThermalBackground(
    n_profile=lambda s: n0 * (1 - s),
    T_profile=lambda s: 10e3 * ONE_EV * (1 - s),
    mass=3 * PROTON_MASS,
    charge=ELEMENTARY_CHARGE,
)
electrons = ThermalBackground(
    n_profile=lambda s: 2 * n0 * (1 - s),   # quasi-neutral
    T_profile=lambda s: 10e3 * ONE_EV * (1 - s),
    mass=ELECTRON_MASS,
    charge=-ELEMENTARY_CHARGE,
)

res_tys, res_hits = trace_particles_boozer_with_collisions(
    field,
    stz_inits,                              # (N, 3) array
    parallel_speeds,                        # (N,) array, m/s
    backgrounds=[deuterium, tritium, electrons],
```

```
    tmax=1e-1,                              # 100 ms
    mass=ALPHA_PARTICLE_MASS,
    charge=ALPHA_PARTICLE_CHARGE,
    Ekin=FUSION_ALPHA_PARTICLE_ENERGY,      # 3.52 MeV
    tol=1e-9,
    dt_save=1e-4,                           # save every 0.1 ms
    DP_hmin=1e-10,                          # s
    rng_seed=42,
)
```

9 Testing and Validation

The implementation is exercised at four levels of increasing scope: analytic unit tests of the coefficient formulae, pure-Python velocity-space relaxation tests with no compiled module or magnetic field, end-to-end tests of the full production tracer (guiding-centre orbits + Milstein noise), and cross-code validation against ASCOT5 [3] in both a controlled uniform plasma and a full reactor-scale 3D stellarator equilibrium. All numbers below are reproducible from `tests/field/test_collisions.py`, `tests/field/test_collisions_local.py`, and the examples referenced in each subsection.

9.1 Analytic Coefficient Unit Tests

`TestCollisionCoefficients` checks the Chandrasekhar function $G(x)$ and its derivative against known limits ($G(0) = 0$, $G(x \to \infty) \to 1/2x^2$, sign change of G' near $x \approx 0.92$), verifies $K < 0$ (drag always decelerates) and $\nu_D > 0$ for arbitrary finite backgrounds, and checks the expected scalings $K \propto n_b$, $K \propto q_a^2$, $\nu_D \propto n_b$, and additivity across species to $\lesssim 10\%$ (the residual comes from the $\ln \Lambda$ dependence on λ_D). `TestUnphysicalCoulombLog` confirms that $\ln \Lambda \leq 0$ raises an exception rather than returning silently unphysical coefficients (Section 4.2). These are deterministic, sub-second checks with no random component.

9.2 Local Velocity-Space Relaxation Tests

`tests/field/test_collisions_local.py` mirrors `collisions.h` in pure Python (no compiled module, no magnetic field) and integrates only the velocity-space SDE (Eqs. (26)–(27) with the orbital drifts set to zero) for an ensemble of $N = 4000$ particles. All collision rates scale linearly with the background density, so boosting $n_b \to 10^{23}\,\mathrm{m}^{-3}$ compresses the relaxation time to microseconds, letting the full suite run in about 10 seconds on a laptop. These tests are the most direct regression check on the Einstein-relation drag (Eq. (16)): an earlier, incorrect form of Q passed all analytic unit tests but drove a proton ensemble to $\langle E \rangle = 3.7\,T_b$ instead of $1.5\,T_b$ here, with a Kolmogorov–Smirnov (KS) p-value of 0 against the Maxwell speed law.

The same ensemble also verifies pitch isotropisation: starting from $\xi_0 = 1$, the relaxed ensemble gives $\langle \xi \rangle = 0.009$ (target 0) and $\langle \xi^2 \rangle = 0.330$ (target 1/3), consistent with convergence to $\xi \sim \mathcal{U}(-1, 1)$. A short-time drift check on 3.52 MeV α-particles against 10 keV electrons, over a window short enough that $K(v)$ is nearly constant (Doppler shift in speed $< 2\%$), gives $\left. \frac{d\langle v \rangle}{dt} \middle/ K(v_0) \right. = 0.984$ (target 1), confirming the deterministic drift term independently of the noise terms.

Test	$\langle E\rangle/T_b$ (target 1.5)	KS p-value	Pass
Cold start ($E_0 = 0.45\,T_b$)	1.446	0.037	✓
Hot start ($E_0 = 4.5\,T_b$)	1.486	0.095	✓
Stationarity (start at equilibrium)	1.500	0.777	✓

Table 2: Relaxation of a proton ensemble in a 1 keV proton background ($n_b = 10^{23}\,\mathrm{m}^{-3}$) to the Maxwellian, evolved for 8 relaxation times. KS p-values test the final speed distribution against the Maxwell speed law at the background temperature.

9.3 End-to-End Equilibration Tests (Production Tracer)

`TestMaxwellianEquilibration` and `TestPitchIsotropization` in `tests/field/test_collisions.py` exercise the full production path — `trace_particles_boozer_with_collisions`, i.e. the adaptive Dormand–Prince orbital integrator coupled to the Milstein noise step, compiled C++ coefficients, real (though simplified, `BoozerAnalytic`) magnetic geometry, and a `MaxToroidalFluxStoppingCriterion` — rather than the idealised velocity-space-only integration of Section 9.2.

Test	Measured	Target	Pass
Hot start ($E_0 = 4.5\,T_b$), $\langle E\rangle/T_b$	1.392	1.5	✓
Cold start ($E_0 = 0.45\,T_b$), $\langle E\rangle/T_b$	1.601	1.5	✓
Pitch isotropisation, $\langle\xi\rangle$	−0.019	0	✓
Pitch isotropisation, $\langle\xi^2\rangle$	0.326	1/3	✓
Pitch broadening ($\xi_0 = 0$), std(ξ)	0.418	> 0.05	✓

Table 3: End-to-end equilibration through the production tracer: 40 protons on a 1 keV proton background, $n_b = 10^{21}\,\mathrm{m}^{-3}$, 8 relaxation times, ITER-scale confining field. All particles remain confined and finite throughout (no losses, no NaN states).

The pitch-decay benchmark and its correction. A quantitative pitch-decay test against the naive expectation $\langle\xi(t)\rangle = \xi_0 e^{-\nu_D t}$ (the usual ASCOT5-style benchmark, which assumes v constant) failed the first time this test suite was run on Perlmutter: for 3.52 MeV α-particles on a single deuterium background, the drag rate is $(m_a/m_b)\,\nu_D = 2\,\nu_D$, so the ensemble slows almost completely over the same window in which pitch decay is measured, and $\nu_D(v) \propto v^{-3}$ diverges as the particles thermalise — the ensemble fully isotropises ($\langle\xi\rangle \to 0$) rather than decaying exponentially at the fixed initial rate. The test was corrected to check the classical relation obtained by eliminating time between the drag and pitch equations,

$$\frac{\langle\xi(t_2)\rangle}{\langle\xi(t_1)\rangle} = \left(\frac{v(t_2)}{v(t_1)}\right)^{m_b/m_a}, \tag{36}$$

evaluated self-consistently against the simulation's own mean speed (referencing the first saved snapshot rather than $t = 0$ cancels a geometric offset from sampling ξ along field-line-following orbits). A second, independent effect was found in the same debugging pass: in a toroidal (non-uniform-$|B|$) field the trapped–passing boundary can sit just below the injected pitch, so scattered particles fall into the trapped population where the orbit-averaged ξ is systematically pulled toward zero, roughly doubling the apparent decay rate relative to the uniform-plasma law — real neoclassical physics, but not what a velocity-space-operator test should measure. The test therefore uses a near-uniform-$|B|$ configuration ($\bar{\eta} \to 0$: no mirror force, no trapped population), giving

$$\frac{\langle \xi_2 \rangle / \langle \xi_1 \rangle}{(v_2/v_1)^{m_b/m_a}} = 1.038 \qquad (\text{target } 1, \ \text{tolerance } \pm 15\%),$$

with $v_2/v_1 = 0.498$ over the measurement window.

9.4 Slow Physics Suite (MPI, HPC)

Five tests marked `@pytest.mark.slow` run longer physical integrations ($N \sim$ 20–128 particles, 5–100 μs of physical time) that are impractical to run serially: an electron-drag signal-to-noise test (chosen specifically because ion-background drag is buried in noise for 3.52 MeV α-particles, whereas electron-background drag is enhanced by m_α/m_e), a cold-plasma energy-loss test, a two-background superposition test, and the pitch-broadening/decay tests of Section 9.3. The tracer's MPI support (particles distributed across ranks, results all-gathered identically on every rank, per-particle seeds rank-independent) lets this suite run on `tests/perlmutter/run_slow_tests.sh`: all five tests pass in 77 s under 64 MPI ranks on Perlmutter, versus 3h45m running the same suite serially.

9.5 Cross-Code Validation Against ASCOT5: Uniform Plasma

This validation mirrors the alpha slowing-down branch of ASCOT5's own built-in Coulomb-collision physics test: N fusion alphas (birth energy 3.52 MeV, isotropic pitch, shared random seed) slow down in a uniform 1 keV hydrogen plasma. The two codes use different magnetic geometries (ASCOT5's analytical ITER-like circular tokamak; firm3d's `BoozerAnalytic` at the same R_0, B_0 scale), but since the plasma is uniform the compared quantities — mean slowing-down time to 50 keV and $\langle E(t) \rangle$ — are geometry-insensitive, isolating the collision operator itself.

	firm3d	ASCOT5	Agreement
Mean slowing-down time [ms]	24.915 ± 0.027	24.885 ± 0.029	+0.1%, 0.7σ

Table 4: $N = 1024$ alphas, $n = 10^{20}\,\mathrm{m}^{-3}$ (the published ASCOT5 test-point density), $T = 1\,\mathrm{keV}$. Also: $\max |\langle E(t) \rangle_{\mathrm{firm3d}} - \langle E(t) \rangle_{\mathrm{ASCOT5}}|/E_0 = 0.4\%$ over the full slowing-down window.

10 Validity and Limitations

- **Vacuum field:** electric fields, $E \times B$ drifts, and electromagnetic perturbations are not included in the collision model (though perturbations can be combined via the existing SAW tracing infrastructure).
- **Maxwellian backgrounds:** the collision operator assumes each background species has a local Maxwellian distribution function. Distorted distributions (e.g. beam-driven) are not supported.

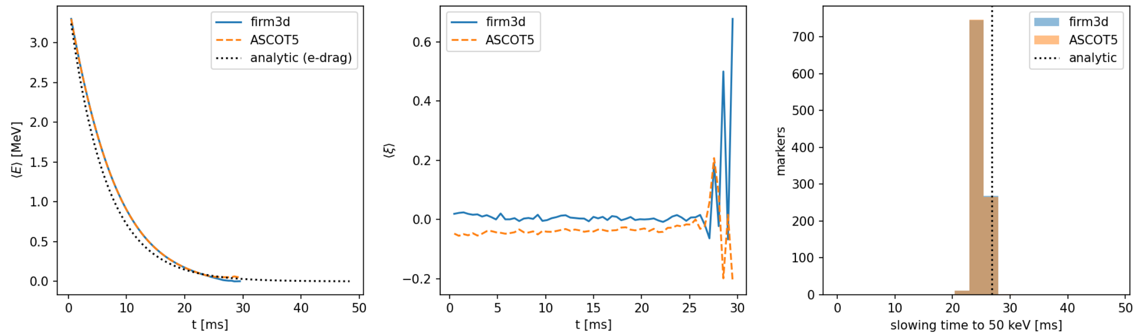

Figure 1: firm3d vs. ASCOT5, uniform 1 keV plasma, $N = 1024$: mean energy $\langle E(t)\rangle$ against the analytic electron-drag estimate $v = v_0 e^{-t/\tau_{se}}$ (left); mean pitch $\langle \xi(t)\rangle$, shown for qualitative comparison only since it is geometry-sensitive (centre); and the distribution of per-particle slowing-down times to 50 keV against the analytic Spitzer estimate (right).

- **Test-particle limit:** the EP density is assumed negligibly small compared to the background, so no back-reaction on the background is included.
- **Guiding-centre approximation:** valid when the Larmor radius $r_L = mv_\perp/(|q|B)$ is small compared to the scale lengths of B and the density/temperature profiles. For 3.52 MeV α-particles in a 5 T stellarator $r_L \sim 5$ cm, comparable to device scale, so finite-Larmor-radius corrections may be non-negligible.
- **Zeroth-order GC collision operator:** the implementation follows Hirvijoki et al. [1] at zeroth order in ρ_*/L (Larmor radius over gradient length). The spatial diffusion term $D_X \propto D_\perp/\Omega_g^2$ present in the full operator [2] is neglected.
- **Coulomb logarithm held fixed in $\partial D_\parallel/\partial v$:** Eq. (15) treats $\ln\Lambda_{ab}$ as constant under differentiation, while the $\ln\Lambda_{ab}$ actually used varies with v through $\bar{v}^2$ (Eq. (9)). The spurious-drift terms in K (Eq. (17)) are then not quite the ones belonging to the $D_\parallel(v)$ that drives the noise, and the stationary solution of the speed SDE becomes the Maxwellian divided by $\ln\Lambda_{ab}(v)$ rather than the Maxwellian itself. Integrating the code's own K and $D_\parallel$ for a proton in 1 keV hydrogen at $n = 10^{21}\,\mathrm{m}^{-3}$ gives $\langle E\rangle/T_b = 1.469$ against the exact 3/2, a -2.1% bias; the closed form above reproduces that number exactly. This is inherited, not accidental: ASCOT5's `mccc_coefs_clog` is likewise velocity-dependent and re-evaluated every step, while `mccc_coefs_dDpara` likewise carries c_{ab} (which contains $\ln\Lambda$) through the derivative, so the two codes agree coefficient for coefficient and share this bias. Adding the $\partial \ln\Lambda_{ab}/\partial v$ term would make the equilibrium exactly Maxwellian at the cost of departing from ASCOT5. Note the effect is on the thermal equilibrium, not on slowing-down: the uniform-plasma benchmark of Section 9.5 agrees with ASCOT5 to 0.7σ, as it should when both codes carry the same approximation.
- **Strong convergence of the noise step:** the diffusion matrix is not commutative, so omitting the Lévy-area terms leaves the scheme strong order 1/2 rather than 1 in the v–ξ coupling; weak order 1 is retained. See Section 5.2.
- **Slow tests: `@pytest.mark.slow`** — five physics-validation tests require $N \sim 20$ particles integrated for 5–100 μs and are intended for HPC resources (Perlmutter or similar).

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